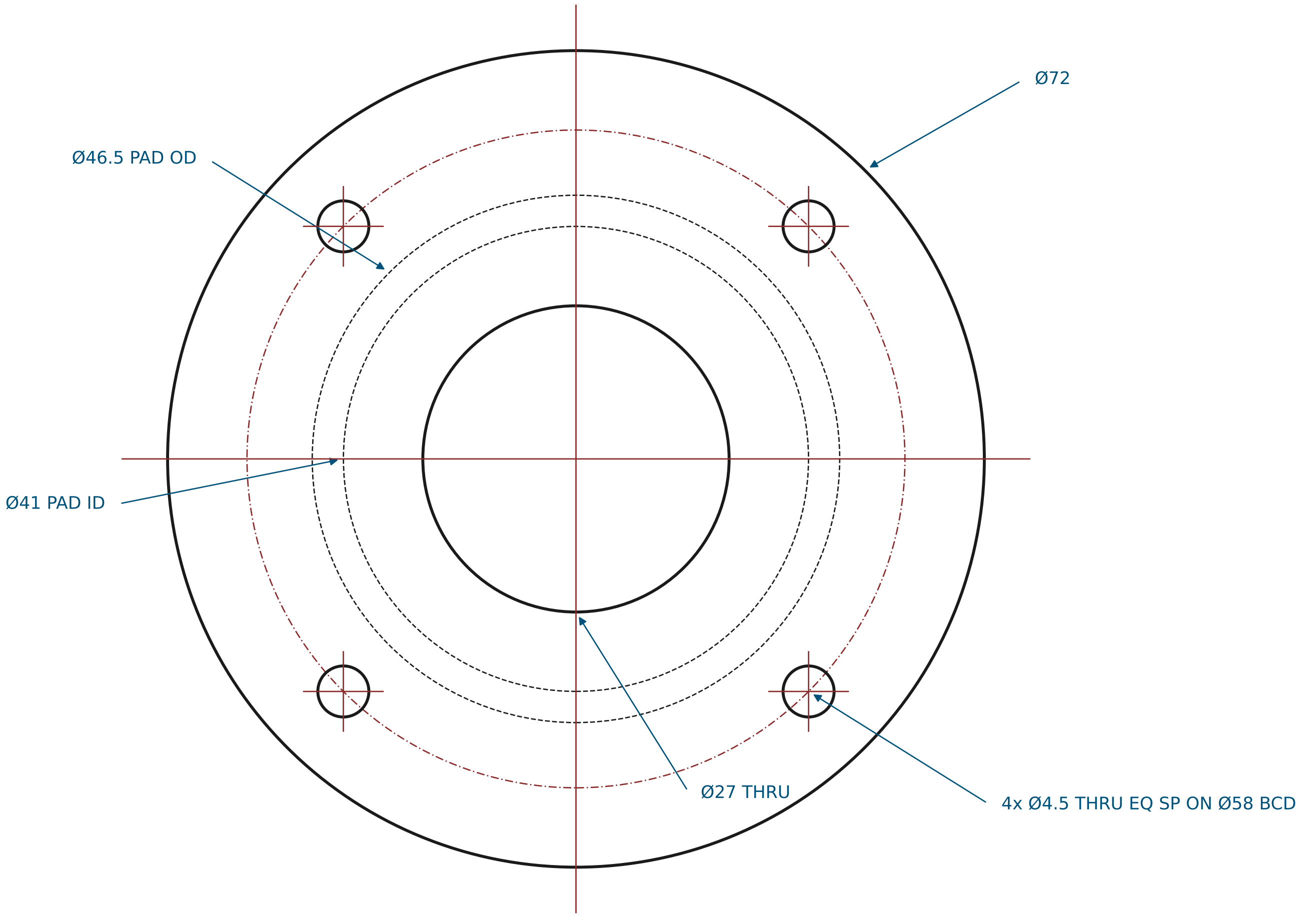
SHEET 1: BEARING RETAINER CAP (BRG-CAP)

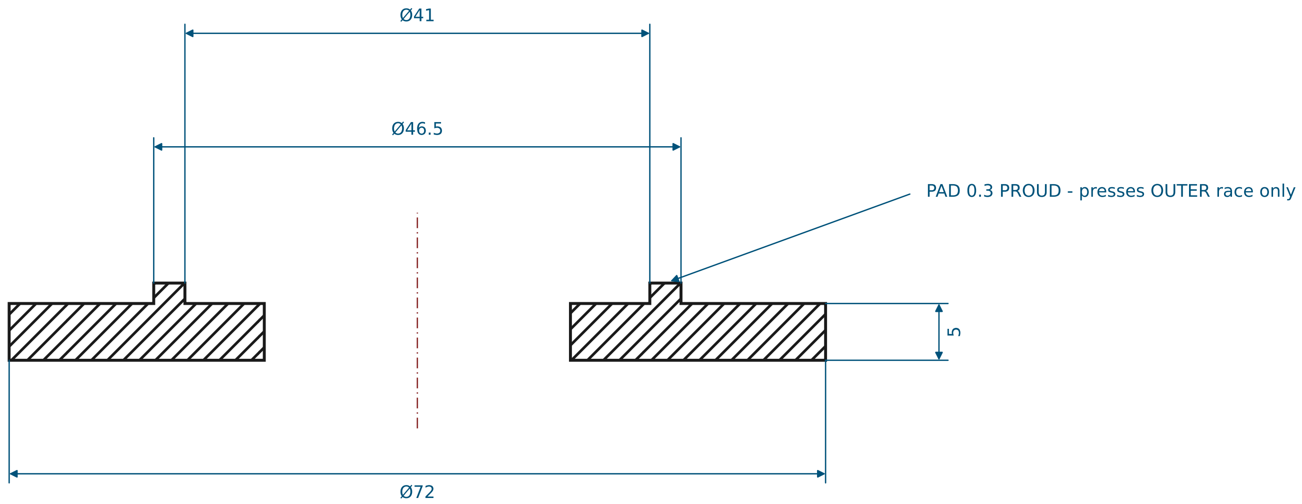
GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

1. One cap above AND one below each 7005 bearing bore; the pair bolts together THROUGH the plate with 4x M4x30 SHCS + nyloc (plate holes 4.5 on the same BCD; grip 22 = 5+12+5 at BOTH plates).
2. Both plates are 12 THK = the 7005 race width, so the race sits FLUSH with the plate faces and the cap disc seats FLAT on the plate.
3. The 0.3-proud pad (41-46.5) lands on the bearing OUTER race only - torquing the bolts gives a small defined clamp crush on the race. No spigot: the cap must NOT stand off the plate.
4. Build one cap pad-UP and one pad-DOWN per bore (same part, flipped).
5. OD is 72 so the bolt holes keep a full 1.5d rim (edge distance 4.75).



PLAN (pad side away)



SECTION A-A (pad height exaggerated)

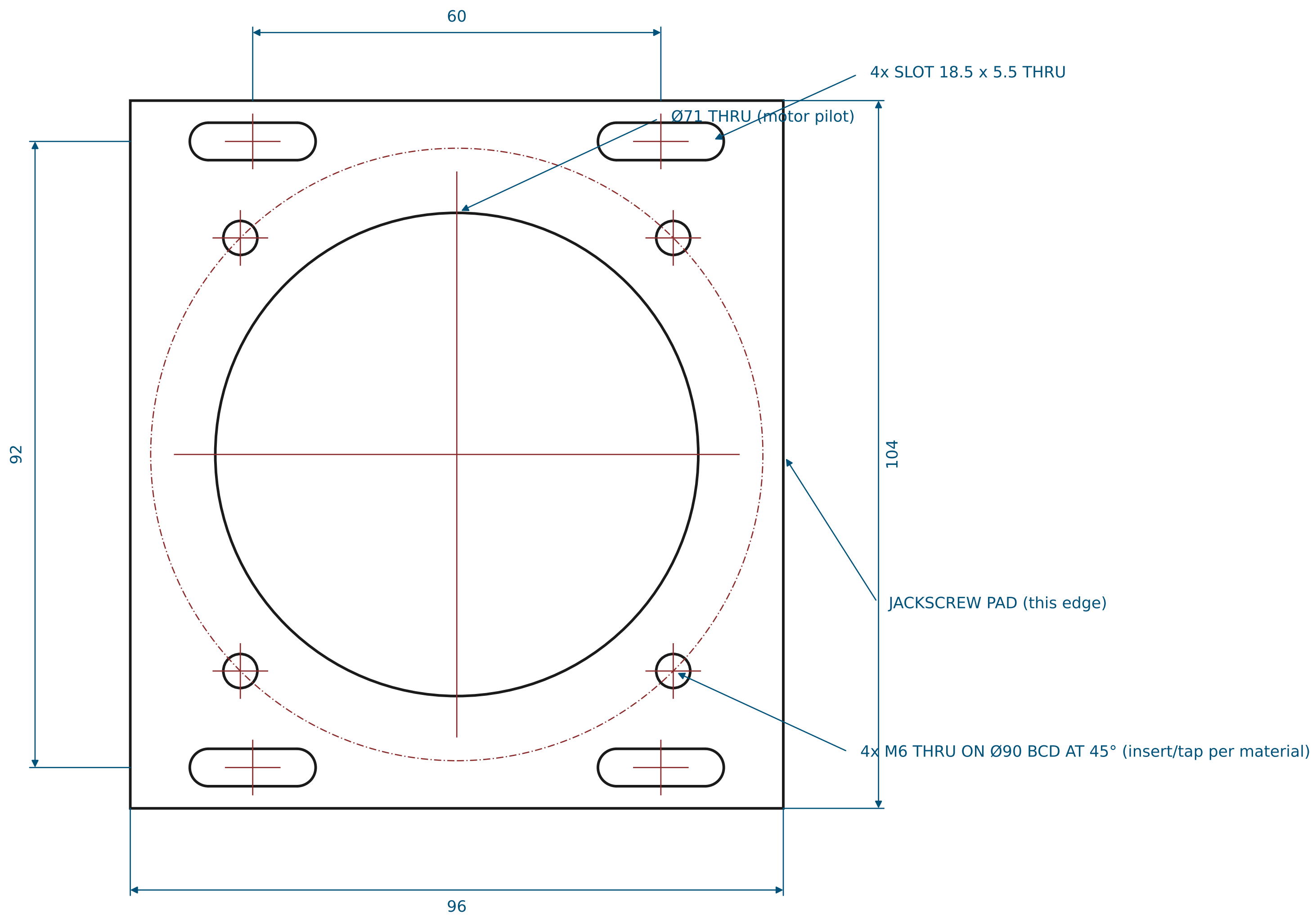
5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: BEARING RETAINER CAP	DWG NO: BCR-01 REV A
MATERIAL: PA12 MJF / 6061-T6	QTY: 8
SCALE 1.5:1 UNITS mm THIRD ANGLE	SHEET 1 OF 6 2026-07-19

SHEET 2: SLIDING MOTOR CARRIAGE (CARRIAGE)

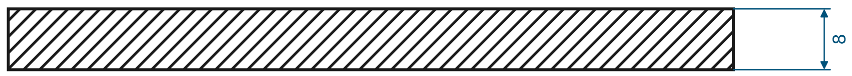
GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

1. Motor (A6M80, 80 sq flange) mounts from BELOW: 4x M6x18 SHCS up through the motor flange into the four M6 positions. PRINTED part: M6 brass heat-set inserts. 6061: tap M6.
2. The four slots take M5x16 SHCS + washer down into the cradle frame bosses directly beneath (tightened from the top through the deck access slots, sheet 4); slot length gives ±6 mm of tension travel.
3. The +X edge (toward the shoulder) is the jackscrew pad — the M6 jackscrew tip bears mid-edge.
4. Same part both sides (symmetric).



PLAN (top face)



EDGE VIEW

5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: SLIDING MOTOR CARRIAGE	DWG NO: BCR-02 REV A
MATERIAL: PA12 MJF / 6061-T6	QTY: 2
SCALE 1:1 UNITS mm THIRD ANGLE	SHEET 2 OF 6 2026-07-19

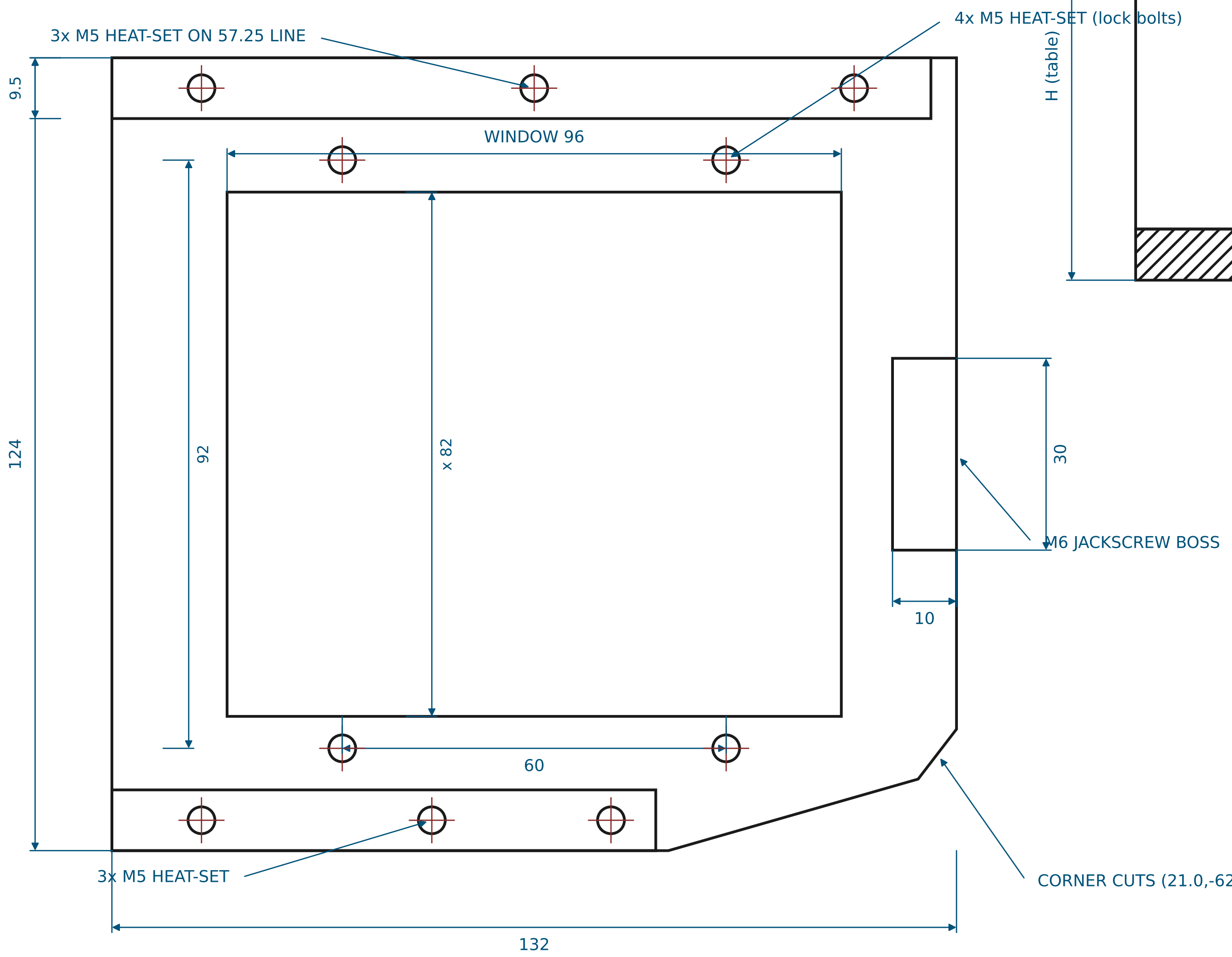
SHEET 3: SLIDER CRADLE (LH SHOWN, RH = MIRROR) (CRADLE-L / CRADLE-R)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

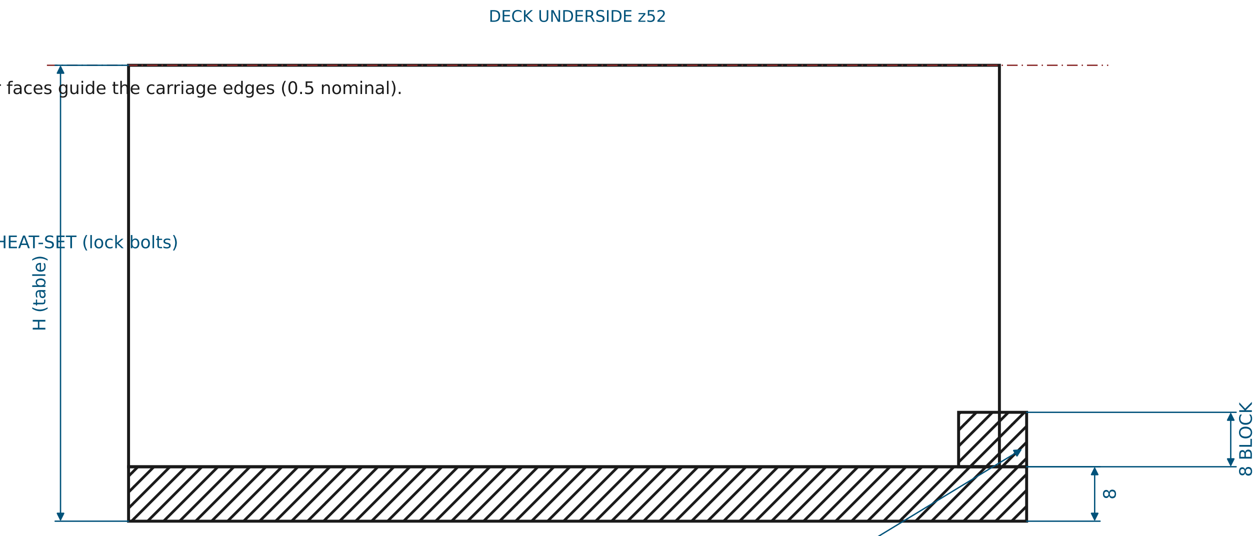
DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

1. ONE printed part per motor = window frame + two shear fins + jack block. The RH part is the exact MIRROR of the LH drawn here; fin/wall heights differ per the table (belt planes stagger).
2. Fins are SOLID full-depth walls (9.5 thick, guide face to frame edge). They bolt UP into the deck underside: 3x M5 full-depth heat-set per fin, M5x20 SHCS from the deck top (deck holes 5.2 THRU).
3. The four M5 positions in the frame border take the carriage lock bolts — fit heat-set inserts.
4. M6 heat-set (or tapped) jackscrew boss in the block; jam nut on the screw against the block face.
5. The chamfered corner keeps the two cradles clear of the machine centreline — do not 'square it up'.
6. FIN WALLS: solid, inner (guide) faces at b = ±52.5, 9.5 thick out to the frame edge. Outboard wall spans a = -66..+62, inboard a = -66..+19. Inner faces guide the carriage edges (0.5 nominal).
7. Install the belt around both pulleys BEFORE bolting the deck down (the fins close the corridor).

PLAN — LH PART (frame top; fins rise toward viewer)



DATUM: origin at frame centre. +a = toward the shoulder (jack-block end), +b = toward the OUTBOARD fin. All holes in the HOLE TABLE.



ELEVATION (outboard fin)

PART TABLE

	FRAME TOP z	WALL H (frame btm→deck)	MIRROR
CRADLE-L	-7	67	as drawn
CRADLE-R	13	47	mirrored

HOLE TABLE (a, b) — LH part; RH = same with b negated

ID	a	b	SPEC
H1	-52.00	57.25	M5 heat-set, fin wall
H2	0.00	57.25	M5 heat-set, fin wall
H3	50.00	57.25	M5 heat-set, fin wall
H4	-52.00	-57.25	M5 heat-set, fin wall
H5	-16.00	-57.25	M5 heat-set, fin wall
H6	12.00	-57.25	M5 heat-set, fin wall
L1	30.00	46.00	M5 heat-set, carriage lock
L2	30.00	-46.00	M5 heat-set, carriage lock
L3	-30.00	46.00	M5 heat-set, carriage lock
L4	-30.00	-46.00	M5 heat-set, carriage lock
J1	61.00	0.00	M6 heat-set, jackscrew

5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: SLIDER CRADLE (LH SHOWN, RH = MIRROR)	DWG NO: BCR-03 REV A
MATERIAL: PA12 MJF / PETG-CF (print flat, frame down)	QTY: 1 + 1 (mirror)
SCALE 1:2 / 1:1 UNITS mm THIRD ANGLE	SHEET 3 OF 6 2026-07-19

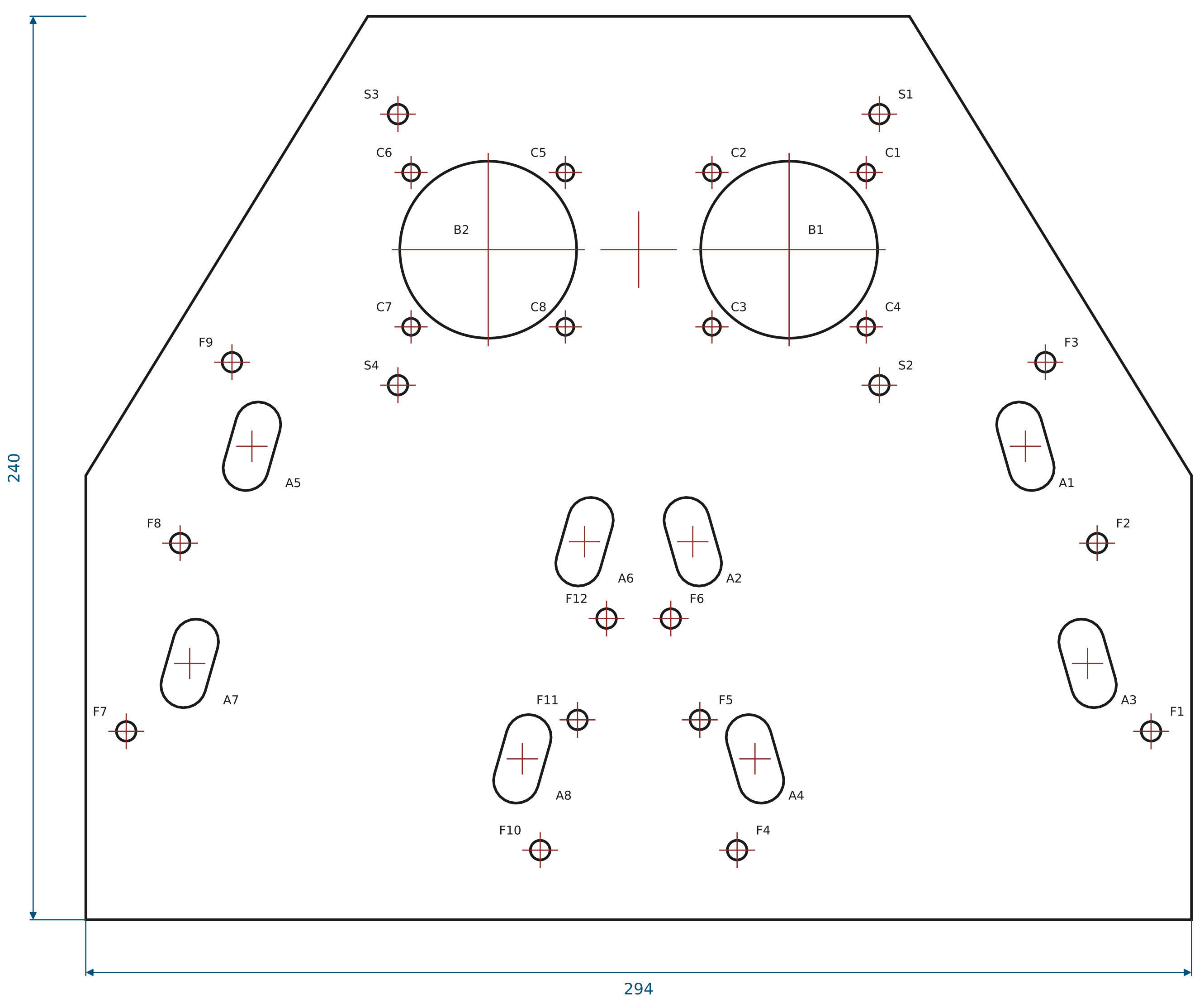
SHEET 4: BOTTOM DECK PLATE (DECK)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

1. Machine per the HOLE TABLE (coordinates from the plate origin = mid-span between the two shoulder bores; +Y toward the arms). All holes THRU.
2. The two 47.0 bores carry the 7005 bearings (transition fit: bore 47.0 +0.025/0).
3. Corner radii R8. Outline vertices in the VERTEX TABLE.
4. 8x ACCESS SLOTS 24 x 12 THRU (table below): allen-key access to the carriage lock bolts ~45 mm below; long axis follows the tension travel.
5. Plate must stay within a 300 x 300 print bed (current 294 x 240).

PLAN — deck top (origin at mid-span of shoulder bores)



HOLE TABLE

ID	X	Y	Ø	SPEC
B1	40.00	0.00	47.0	7005 bearing bore, thru
B2	-40.00	0.00	47.0	7005 bearing bore, thru
C1	60.51	20.51	4.5	M4 cap bolt, thru
C2	19.49	20.51	4.5	M4 cap bolt, thru
C3	19.49	-20.51	4.5	M4 cap bolt, thru
C4	60.51	-20.51	4.5	M4 cap bolt, thru
C5	-19.49	20.51	4.5	M4 cap bolt, thru
C6	-60.51	20.51	4.5	M4 cap bolt, thru
C7	-60.51	-20.51	4.5	M4 cap bolt, thru
C8	-19.49	-20.51	4.5	M4 cap bolt, thru
S1	64.00	36.00	5.2	M5 standoff bolt, thru
S2	64.00	-36.00	5.2	M5 standoff bolt, thru
S3	-64.00	36.00	5.2	M5 standoff bolt, thru
S4	-64.00	-36.00	5.2	M5 standoff bolt, thru
F1	136.25	-127.95	5.2	M5 fin bolt, thru
F2	121.91	-77.96	5.2	M5 fin bolt, thru
F3	108.13	-29.90	5.2	M5 fin bolt, thru
F4	26.18	-159.51	5.2	M5 fin bolt, thru
F5	16.26	-124.90	5.2	M5 fin bolt, thru
F6	8.54	-97.99	5.2	M5 fin bolt, thru
F7	-136.25	-127.95	5.2	M5 fin bolt, thru
F8	-121.91	-77.96	5.2	M5 fin bolt, thru
F9	-108.13	-29.90	5.2	M5 fin bolt, thru
F10	-26.18	-159.51	5.2	M5 fin bolt, thru
F11	-16.26	-124.90	5.2	M5 fin bolt, thru
F12	-8.54	-97.99	5.2	M5 fin bolt, thru

ACCESS SLOTS 24 x 12 THRU (X, Y, LONG-AXIS ANGLE)

A1	102.83	-52.22	106.00 deg	A2	14.39	-77.58	106.00 deg
A3	119.37	-109.90	106.00 deg	A4	30.93	-135.26	106.00 deg
A5	-102.83	-52.22	74.00 deg	A6	-14.39	-77.58	74.00 deg
A7	-119.37	-109.90	74.00 deg	A8	-30.93	-135.26	74.00 deg

VERTEX TABLE (outline, R8 corners)

V1 (-72,62)	V2 (72,62)	V3 (147,-60)
V4 (147,-178)	V5 (-147,-178)	V6 (-147,-60)

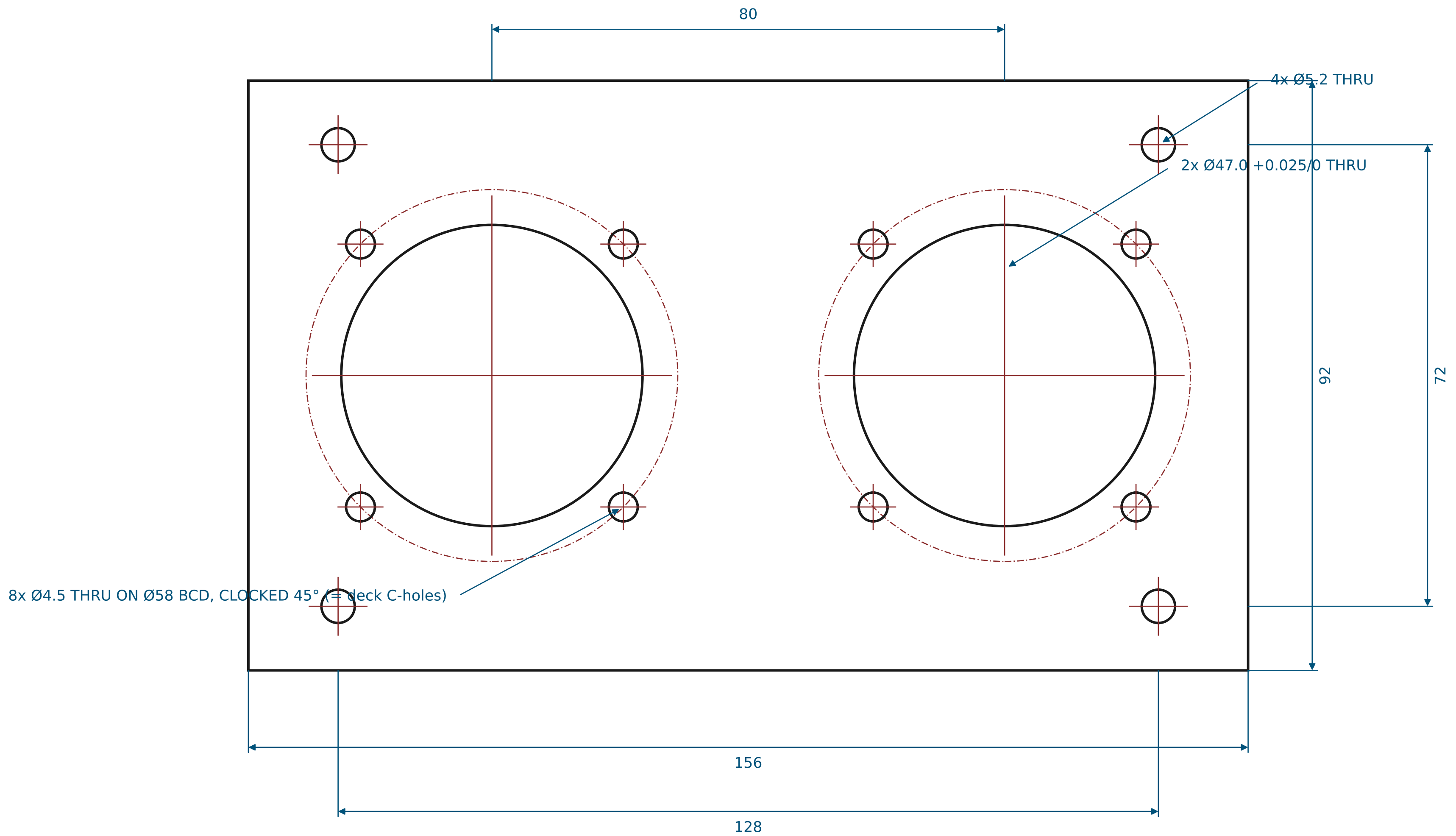
5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: BOTTOM DECK PLATE	DWG NO: BCR-04 REV A
MATERIAL: PA12 MJF / 6061-T6 (12 THK)	QTY: 1
SCALE 1:2 UNITS mm THIRD ANGLE	SHEET 4 OF 6 2026-07-19

SHEET 5: TOP PLATE + STANDOFFS   (TOP-PLATE)

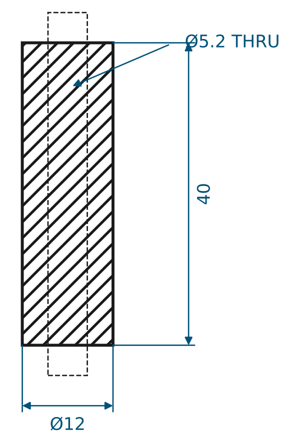
GENERAL TOLERANCES UNLESS NOTED: X ±0.5   X.X ±0.2   HOLES +0.2/-0   ANGLES ±0.5°   BREAK ALL EDGES 0.3

DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

1. Same bearing bore + cap-bolt pattern as the deck (47.0 bores, 4x 4.5 on Ø58 BCD per bore). 12 THK = the 7005 race width: the race sits flush with both faces (flat caps seat on the plate).
2. Standoff bolt holes moved OUT to (±64, ±36) so the Ø72 caps clear the standoff bosses (they collided with the old Ø62 caps at ±60,±32).
3. 4x standoffs Ø12 x 40, Ø5.2 clearance bore THRU. Each stack is THROUGH-BOLTED: one M5 x 75 SHCS from the plate top, nyloc + washer under the deck (no threads in the standoff; grip 64).



PLAN



STANDOFF (x4)

5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: TOP PLATE + STANDOFFS	DWG NO: BCR-05 REV A
MATERIAL: PA12 MJF / 6061-T6 (12 THK = 7005 race width)	QTY: 1 (+4 standoffs)
SCALE 1:1 UNITS mm THIRD ANGLE	SHEET 5 OF 6 2026-07-19

SHEET 6: TENSIONER ASSEMBLY + HARDWARE   (ASSY-TENSIONER)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5   X.X ±0.2   HOLES +0.2/-0   ANGLES ±0.5°   BREAK ALL EDGES 0.3

DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

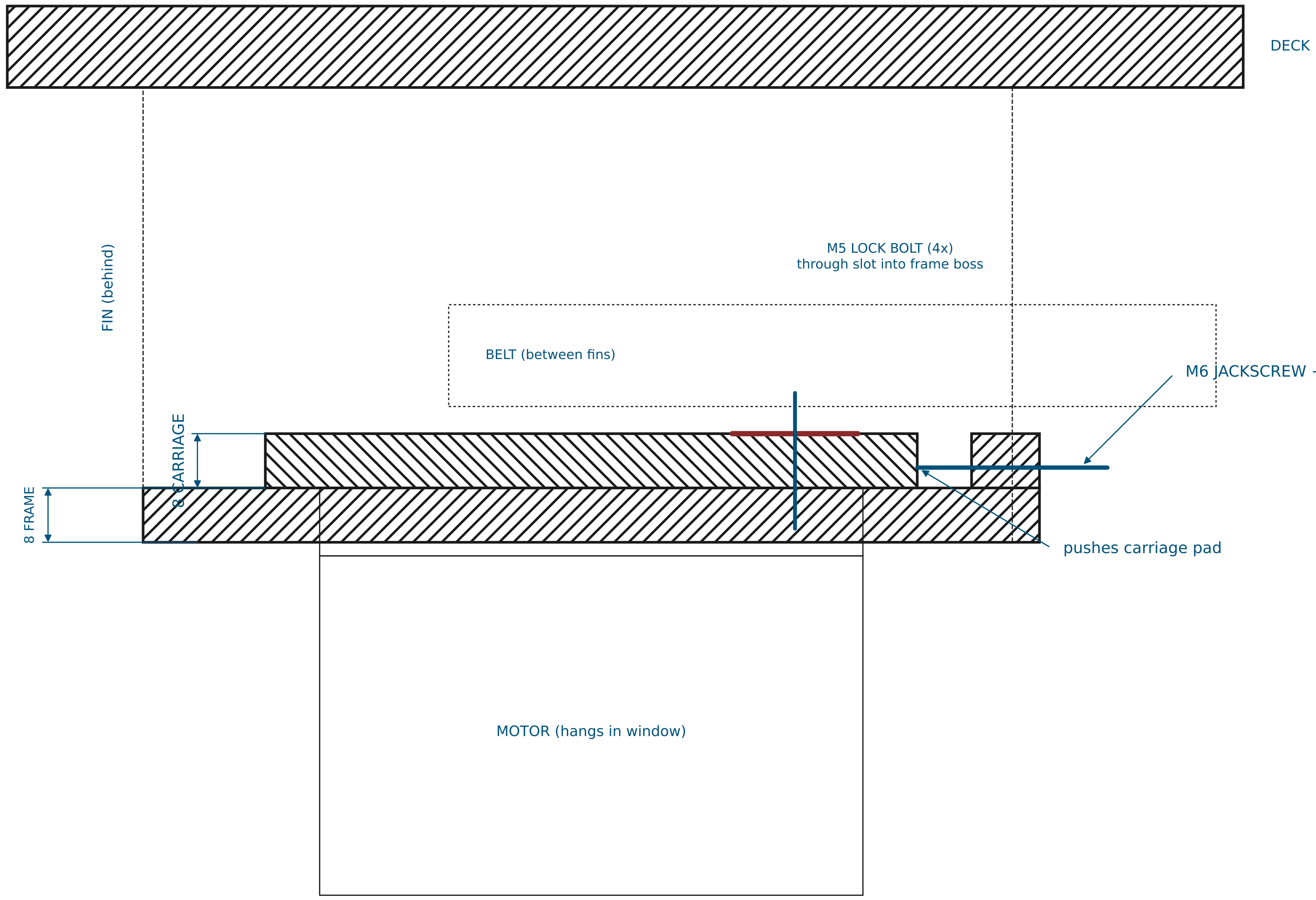
1. TENSION PROCEDURE: loosen the 4x M5 lock bolts 1/2 turn (allen key from the TOP, through the deck access slots) -> turn the M6 jackscrew CW to push the carriage away from the shoulder (belt tightens) -> torque the lock bolts (M5: 4 N·m in inserts) -> set the M6 jam nut against the block.

2. Travel available ±6 mm. Belt: 450-5M-15 (24T -> 72T, C = 97.5 nominal).

3. ASSEMBLY ORDER: pulleys + belt on motor/shaft FIRST, then lower the deck onto fins + standoffs and bolt down (the fin walls close the belt corridor once the deck is on).

4. The belt passes BETWEEN the two fins, above the frame and below the deck — nothing crosses it.

SECTION ON THE BELT AXIS — LEFT SIDE (right side mirrors, +20 in z)



HARDWARE (per machine)

```
16x  M4 x 30 SHCS + nyloc + washers
     bearing cap pairs, 4 per bore stack (grip 22 both plates)
8x   M5 x 16 SHCS + washer
     carriage lock bolts (4 per side)
12x  M5 x 20 SHCS
     fin walls -> deck underside (full-depth heat-sets)
4x   M5 x 75 SHCS + nyloc
     top plate -> standoff -> deck through-bolt (one per standoff, grip 64)
8x   M6 x 18 SHCS
     motor flange -> carriage from below (full 8 mm engagement)
2x   M6 x 40 jackscrew + jam nut
     tensioner
2x   M25x1.5 shaft locknut (KM5)
     bearing preload
--   M5/M6 brass heat-set inserts
     all printed bosses
```

5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: TENSIONER ASSEMBLY + HARDWARE	DWG NO: BCR-06 REV A
MATERIAL: -	QTY: 2 (mirrored)
SCALE 1:1 section UNITS mm THIRD ANGLE	SHEET 6 OF 6 2026-07-19